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SGQR-RAG: A Structure-Guided Retrieval Framework for Multi-Turn Question Answering in Aviation Manuals

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Abstract:	In complex engineering documents such as aviation manuals, multi-turn question answering often suffers from incomplete query semantics, high sensitivity to domain-specific terminology, and insufficient use of document structural information. These issues make it difficult for retrieval-augmented generation (RAG) methods to consistently produce high-quality answers. To address these challenges, this paper proposes SGQR-RAG, a structure-guided retrieval framework for multi-turn question answering. First, we construct a multi-turn QA dataset for aviation manuals and characterize the degree of query dependence on context using the Information Hiding Rate (IHR), so as to reflect task difficulty. We then propose SGQR, a structure-guided query rewriting method. In the first stage, query completeness is restored through completion-based query rewriting with keyword preservation constraints. In the second stage, a gated path expansion strategy is introduced based on the title distribution entropy of initial retrieval results, which incorporates high-confidence structural information while suppressing structural noise. Furthermore, we adopt a hybrid retrieval strategy that integrates title BM25, context BM25, and dense retrieval, together with answer-coverage-based reranking to optimize evidence organization for generation. To better evaluate the ability of retrieved results to jointly support answer generation, we introduce Cumulative Answer Relevance (CAR), a metric designed to measure the supportiveness of retrieval results for answer generation. Experimental results show that the proposed method achieves consistent improvements across different dialogue scenarios and multiple open-source large language models, outperforming baseline methods in terms of MRR, CAR, and answer quality metrics. These findings verify the effectiveness of SGQR-RAG for multi-turn question answering over aviation manuals and highlight the importance of structural information and answer-coverage modeling.

Dear Editor:

We are pleased to submit our manuscript entitled “SGQR-RAG: A Structure-Guided Retrieval Framework for Multi-Turn Question Answering in Aviation Manuals” for consideration for publication in *Advanced Engineering Informatics*. The manuscript was authored by Yitian Wang, Zixuan Wang, Liang Chen, Aihua Su, and Yu Zheng.

This manuscript presents a structure-guided retrieval framework for multi-turn question answering in complex engineering documents, with a particular focus on aviation manuals. Specifically, the paper addresses three key challenges in multi-turn retrieval-augmented generation settings, including incomplete query semantics, sensitivity to domain-specific terminology, and insufficient use of document structural information. To tackle these issues, the manuscript proposes SGQR-RAG, which integrates completion-based query rewriting with keyword preservation, entropy-gated structural expansion, hybrid retrieval, and answer-coverage-based reranking. In addition, the paper constructs a multi-turn QA dataset based on aviation manuals and introduces the Information Hiding Rate (IHR) and Cumulative Answer Relevance (CAR) to analyze dataset difficulty and retrieval effectiveness. Experimental results across multiple dialogue scenarios and several open-source large language models demonstrate that the proposed framework consistently improves retrieval quality and answer quality.

Further, we believe that this manuscript will be of interest to the readership of this special issue because it studies the application of large language models to question answering over aviation manuals. As aviation manuals are a core source of technical knowledge in aircraft engineering and operations, the proposed framework may provide useful insights into how large language models can better support reliable knowledge access in aviation contexts. In addition, the paper presents an integrated approach that combines query rewriting, retrieval, and evidence reranking, which may also be of interest to researchers working on technical-document understanding and intelligent support for aviation-related tasks.

This manuscript has not been published or presented elsewhere in part or in entirety and is not under consideration by another journal. We have read and understood your journal’s policies, and we believe that neither the manuscript nor the study violates any of these. There are no conflicts of interest to declare.

Thank you for your consideration. I look forward to hearing from you.

Sincerely,

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SGQR-RAG: A Structure-Guided Retrieval Framework for Multi-Turn Question Answering in Aviation Manuals

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Abstract: In complex engineering documents such as aviation manuals, multi-turn question answering often suffers from incomplete query semantics, high sensitivity to domain-specific terminology, and insufficient use of document structural information. These issues make it difficult for retrieval-augmented generation (RAG) methods to consistently produce high-quality answers. To address these challenges, this paper proposes SGQR-RAG, a structure-guided retrieval framework for multi-turn question answering. First, we construct a multi-turn QA dataset for aviation manuals and characterize the degree of query dependence on context using the Information Hiding Rate (IHR), so as to reflect task difficulty. We then propose SGQR, a structure-guided query rewriting method. In the first stage, query completeness is restored through completion-based query rewriting with keyword preservation constraints. In the second stage, a gated path expansion strategy is introduced based on the title distribution entropy of initial retrieval results, which incorporates high-confidence structural information while suppressing structural noise. Furthermore, we adopt a hybrid retrieval strategy that integrates title BM25, context BM25, and dense retrieval, together with answer-coverage-based reranking to optimize evidence organization for generation. To better evaluate the ability of retrieved results to jointly support answer generation, we introduce Cumulative Answer Relevance (CAR), a metric designed to measure the supportiveness of retrieval results for answer generation. Experimental results show that the proposed method achieves consistent improvements across different dialogue scenarios and multiple open-source large language models, outperforming baseline methods in terms of MRR, CAR, and answer quality metrics. These findings verify the effectiveness of SGQR-RAG for multi-turn question answering over aviation manuals and highlight the importance of structural information and answer-coverage modeling.

Keywords: Retrieval-augmented generation; multi-turn question answering; structure-guided query rewriting; aviation manuals; large language models

1 Introduction

In the aviation domain, operational decision-making relies heavily on complex technical documents and professional knowledge systems, such as flight manuals, system descriptions, and maintenance specifications. These documents are typically characterized by clear

hierarchical structures, dense domain-specific terminology, and strong contextual dependencies. In real-world applications, operators often need to locate critical information and make decisions rapidly in environments involving multitasking and high cognitive load [1]-[3]. However, traditional methods based on keyword retrieval or rule matching are not well suited to complex semantic understanding or multi-turn interaction [4], which limits their applicability in intelligent assistance systems.

In recent years, with the rapid development of large language models (LLMs), their strong capabilities in natural language understanding and generation have shown great potential for building intelligent question answering and decision support systems. Under the paradigm of retrieval-augmented generation (RAG) [5],[6], LLMs can compensate for the limitations of parametric knowledge through external knowledge retrieval, thereby improving the accuracy and reliability of generated content. However, in multi-turn dialogue settings, existing RAG methods still face several challenges. On the one hand, multi-turn queries often contain incomplete information and ambiguous references [7],[8], which makes it difficult for the retrieval stage to accurately match the target documents. On the other hand, the structural information embedded in aviation documents, such as chapter hierarchies and system-level organization, is not fully utilized, making it hard for the retrieval process to align with the way professional knowledge is organized. In addition, most existing methods optimize mainly for relevance matching, typically improving retrieval accuracy through reranking models [9],[10], while paying relatively limited attention to the actual usefulness of retrieved results for downstream answer generation.

To address these issues, this paper proposes SGQR-RAG, a query rewriting and retrieval optimization method for multi-turn RAG. The method systematically improves the conventional RAG pipeline from two key perspectives: query construction and retrieval objectives. Specifically, on the query side, we propose Structure-Guided Query Rewriting (SGQR), which refines multi-turn dialogue queries through semantic completion and structural information integration. On the retrieval side, we further introduce Answer-Coverage-Based Reranking, so that retrieved results can better support downstream answer generation, thereby improving overall question answering quality.

The main contributions are as follows:

(1) We propose SGQR, a query rewriting method for multi-turn RAG. To address incomplete query information, strong semantic dependence on dialogue history, and irregular query expression in multi-turn conversations, the method introduces a semantic completion mechanism with keyword preservation constraints and further incorporates document structural information for guided expansion. This enables a more refined reconstruction of the original query, improving both query completeness and retrieval effectiveness.

(2) We develop the SGQR-RAG framework, which enables collaborative design from query rewriting to retrieval optimization. Within this framework, structure-aware query rewriting is combined with Answer-Coverage-Based Reranking, allowing retrieval results to better support downstream generation and shifting retrieval optimization from relevance matching toward answer-coverage-aware evidence organization.

(3) We construct a multi-turn QA dataset in the aviation manual scenario and use Information Hiding Rate (IHR) and Cumulative Answer Relevance (CAR) to analyze dataset difficulty and retrieval effectiveness, respectively. Experimental results across different

dialogue scenarios and multiple open-source LLMs show that the proposed method achieves consistent improvements in MRR, CAR, and answer quality metrics, demonstrating both its effectiveness and generalizability.

2 Related Work

2.1 RAG in the Aviation Domain

In recent years, as large language models have been increasingly applied to domain-specific knowledge question answering, retrieval-augmented generation (RAG) has gradually entered the aviation domain. Existing studies show that RAG applications in aviation have expanded from early domain-specific language models and knowledge question answering to multiple directions, including pilot support, fault diagnosis, accident analysis, and engineering design. Overall, these studies consistently indicate that aviation knowledge is characterized by complex documents, dense terminology, and clear structural hierarchies. They also suggest that relying solely on the parametric memory of large language models is insufficient for highly reliable question answering, making the integration of external knowledge retrieval a natural technical solution.

The studies most closely related to this work are those on pilot support and aviation knowledge question answering. Early work explored the use of natural language question answering systems to directly access pilot documents, with the goal of replacing traditional keyword-based retrieval and improving the usability of complex technical documentation [11]. Subsequent studies gradually introduced domain knowledge modeling and retrieval-augmented mechanisms into aviation applications. For example, knowledge graphs, structured knowledge, and contextual retrieval have been used to improve question understanding and answer generation [12],[13]. AviationCopilot [14] further integrates structured data construction, knowledge-alignment training, and retrieval-based context augmentation to build a relatively complete aviation copilot framework for flight operation assistance. This work represents a broader trend in the field, moving from document question answering toward task-level flight support.

Beyond pilot support, RAG has also developed rapidly in aviation fault diagnosis and maintenance. Related studies typically combine knowledge graphs, fault mechanism modeling, and large language models to support equipment fault localization and maintenance decision-making through graph retrieval, text retrieval, and generative reasoning [15],[16]. For example, Li et al. [17] combined knowledge-graph-driven fault representation with large language models for aviation equipment fault diagnosis. HybridRAG [18], in contrast, integrates graph-structured reasoning, sparse retrieval, and vector retrieval for aircraft troubleshooting, thereby enhancing diagnostic capability in complex maintenance scenarios.

In the area of aviation accident analysis, RAG has further been used to identify key causal relationships from accident reports and support safety analysis. A representative example is RACI [19], which applies retrieval-augmented generation to extract causal information from historical accident reports. Subsequent work has further combined knowledge graphs with prompt engineering to conduct structured analysis of accident causal chains [20].

In addition, RAG has begun to support knowledge-intensive tasks in aviation engineering design and manufacturing. For instance, DR-RAG [21] combines domain rules, knowledge

graphs, and RAG for design assistance in aviation manufacturing scenarios. Some studies have also applied RAG to aircraft design knowledge question answering and requirements engineering support [22]. These studies further demonstrate that RAG in the aviation domain is gradually extending from standalone question answering to more complex engineering decision-making tasks.

Overall, existing RAG research in the aviation domain has covered multiple scenarios, including flight support, maintenance diagnosis, accident analysis, and engineering design, thereby validating the value of retrieval augmentation in professional aviation knowledge applications. However, most prior work has focused on single-turn queries, knowledge enhancement, or reasoning for specific tasks, while paying limited attention to how query semantics can be stably completed in multi-turn dialogue settings, how strong contextual dependencies can be handled, and how structural information can be effectively incorporated into the query formulation process.

2.2 Multi-Turn Dialogue in Closed Domains

In closed-domain settings, research on multi-turn QA is often closely related to document-grounded dialogue, whose core task is to continuously interpret the incomplete question in the current turn based on a given document and the dialogue history, and to locate textual evidence that can support the answer. Compared with open-domain QA, this task places greater emphasis on the precise use of the given documents and on the joint modeling of dialogue context and document content. Existing studies have first advanced this direction from the perspectives of dataset construction and task formulation. For example, Feng et al. [23] first proposed the doc2dial dataset for goal-oriented document-grounded dialogue, which combines multi-turn interaction with document evidence grounding and explicitly defines the task setting of using external documents for knowledge support during dialogue. Building on this, Feng et al. further proposed MultiDoc2Dial [24], which extends the problem to multi-document scenarios and shows that real-world information-seeking dialogues often require knowledge localization and selection across multiple relevant documents. Meanwhile, Ma et al. [25] provided a systematic survey of document-grounded dialogue systems, reviewing this research line from task definition to knowledge selection and response generation, and further highlighting that multi-turn context modeling and document alignment are core challenges in this field.

At the methodological level, existing studies on closed-domain multi-turn dialogue mainly focus on two components: knowledge grounding and response generation. One line of work adopts a pipeline framework, where the system first identifies the evidence passages most relevant to the current turn from the given documents and then generates a response based on them. Another line of work attempts to jointly model knowledge selection and answer generation in order to reduce the effect of error propagation. For example, the DialDoc shared task explicitly defines knowledge grounding and response generation as the key evaluation components in document-grounded multi-turn dialogue, systematically reflecting the basic paradigm of first locating evidence and then generating a response in closed-domain multi-turn dialogue [26]. Along this line, Daheim et al. [27] proposed a cascaded approach that first extracts supporting passages from documents and then generates responses based on the selected evidence, representing a typical two-stage modeling strategy. At the same time, Gao et al. [28] further explored a unified generation framework that jointly models knowledge

grounding and response generation, with the aim of improving consistency between document evidence utilization and response generation. In addition, regarding knowledge selection in multi-turn dialogue, Kim et al. [29] improved evidence grounding and response generation in document-grounded dialogue by designing diverse input representations for pretrained models. More recently, Roy et al. [30] further pointed out from the perspective of multi-turn RAG that, in multi-turn settings, a system must not only decide when to retrieve, but also summarize the current dialogue into a query formulation that is more suitable for retrieval. These studies suggest that, in closed-domain multi-turn dialogue, the quality of current-turn question understanding directly affects both knowledge grounding and response generation. How to transform context-dependent questions into more stable, clearer, and retrievable query expressions therefore remains an important research problem.

Although substantial progress has been made in closed-domain multi-turn dialogue, most existing work focuses primarily on response generation or knowledge snippet selection, while paying relatively limited attention to how context-dependent questions in the current turn can be transformed into queries that are easier to retrieve from. This issue is particularly important for professional documents with explicit hierarchical structure, where current-turn questions often involve omitted terms, referential expressions, and topic continuation. In such cases, context modeling based only on dialogue history concatenation or dialogue-state encoding is still insufficient to fully recover query semantics and align them with document structure. Therefore, it is necessary to further study query reformulation in multi-turn dialogue, which naturally motivates the query rewriting methods discussed in the next section.

2.3 Query Rewriting

In multi-turn dialogue settings, query rewriting aims to reformulate the current context-dependent question into a semantically complete and self-contained query that can be directly processed by downstream retrieval or question answering modules. Early studies in this area established the task definition and its basic paradigm. For example, Elgohary et al. [8] systematized the task of rewriting context-dependent questions into standalone questions through the CANARD dataset, which helped establish the research direction of conversational query rewriting. Subsequently, Vakulenko et al. [7] further decomposed conversational question answering into two sub-tasks, namely query rewriting and question answering, showing that rewriting can serve as an intermediate step connecting contextual understanding with downstream question answering modules. Meanwhile, Anantha et al. [31] proposed QReCC, which unified rewriting, retrieval, and reading comprehension within an open-retrieval conversational QA framework, further demonstrating the foundational role of query rewriting in end-to-end conversational question answering.

Building on this line of work, later studies gradually shifted from simply imitating human rewrites to optimizing query rewriting for downstream retrieval. One representative line of research directly leverages retrieval feedback to train rewriting models. Wu et al. [32] proposed CONQRR, which uses reinforcement learning with retrieval results as reward signals, so that rewritten queries can better adapt to a fixed black-box retriever and thereby improve retrieval performance. Qian et al. [33] investigated explicit query rewriting for conversational dense retrieval, showing that rewriting is not only applicable to traditional retrieval settings but can also be closely integrated with conversational dense retrieval. Mao et al. [34] further proposed

EDIRCS, which adopts a text-editing-based rewriting strategy and introduces training objectives oriented toward search effectiveness, thereby reducing the inefficiency and bias associated with generating rewrites from scratch. Taken together, these studies suggest that the objective of query rewriting has gradually shifted from producing formally complete questions to producing retrieval-oriented queries.

With the development of large language models, research on query rewriting has further expanded toward using LLMs for history disambiguation and feedback alignment. Mo et al. [35] proposed CHIQ, a two-stage method that first enhances ambiguities in the dialogue history and then performs query rewriting, with a particular focus on the role of open-source LLMs in conversational query rewriting. Zhang et al. [36] proposed AdaQR, which constructs pairwise preference signals among rewrite candidates based on retriever preferences and aligns the rewriting model through preference optimization, thereby improving rewrite quality without passage-level annotations. Mao et al. [37] further proposed RaFe, which uses reranker feedback to provide training signals for query rewriting that are more consistent with downstream ranking objectives, thus reducing reliance on manual annotation and handcrafted reward design. These more recent studies suggest that query rewriting should no longer be treated as an isolated reformulation problem, but rather as a component that needs to be optimized together with retrieval and generation.

Overall, existing research on query rewriting has gradually evolved from early supervised standalone rewriting to multiple paradigms oriented toward retrieval feedback, text-editing-based optimization, and LLM-assisted rewriting. Prior work has demonstrated the effectiveness of query rewriting in alleviating omission, reference, and semantic continuation issues in multi-turn dialogue. However, most existing methods still focus mainly on generating more complete queries or improving downstream retrieval performance. Relatively limited attention has been paid to how the hierarchical structural information in professional documents can be stably utilized, and how evidence can be further organized in a way that better supports downstream generation.

3 Methodology

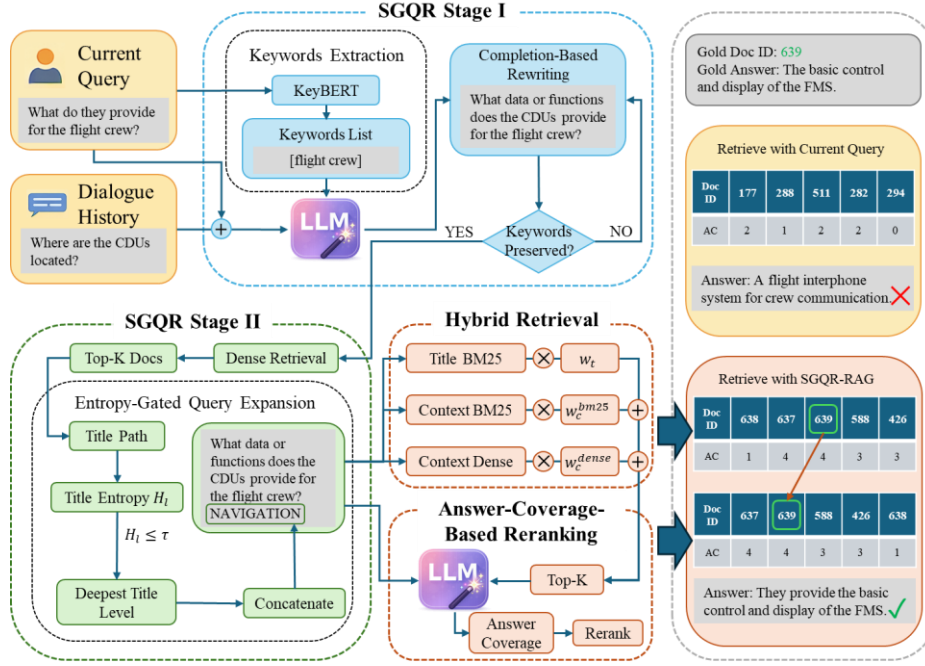


Fig. 1. Overview of the SGQR-RAG framework with an illustrative example. Stage I performs completion-based rewriting with keyword preservation, while Stage II conducts entropy-gated query expansion. The expanded query is then processed by hybrid retrieval and answer-coverage-based reranking, leading to improved evidence selection and answer quality.

3.1 SGQR

The goal of SGQR is to transform the original query in the current turn, which is often semantically incomplete and dependent on dialogue history, into a structure-aware query that is better suited for retrieval in multi-turn dialogue settings. Unlike conventional query rewriting methods, SGQR does not treat rewriting as a purely semantic reformulation problem. Instead, it decomposes the process into two consecutive stages. In the first stage, the complete semantics of the current question are recovered through completion-based rewriting, while keyword preservation is imposed to prevent the loss of domain-specific terminology. In the second stage, structural information from the document is introduced based on initial retrieval results to perform controlled query expansion, thereby further improving the alignment between the query and the target documents.

3.1.1 Stage I: Completion-Based Query Rewriting with Keyword Preservation

In multi-turn dialogue, the current query often depends on dialogue history and typically contains references, omissions, and abbreviated technical terms, making it difficult to express complete semantics independently. To address this issue, the first stage of SGQR performs completion-based rewriting on the current query so that it becomes complete, clear, and directly retrievable.

In professional technical documents such as aviation manuals, user questions are usually centered on specific systems, functional modules, or domain-specific terms, such as *VNAV* and *Autoflight System*. These terms are highly discriminative for retrieval. In our experiments, we

found that although large language models are capable of semantic completion, they may also omit, replace, or over-generalize key terms during rewriting. This can lead to semantic drift and weaken retrieval effectiveness.

To mitigate this problem, we introduce a keyword preservation mechanism into completion-based rewriting to enhance terminology stability. Specifically, we first use KeyBERT [38] to extract highly relevant keywords from the current query, covering domain terms, entity names, and functional descriptors. These keywords are then incorporated into the rewriting prompt to constrain the model to preserve the core terms or their equivalent expressions while completing the missing semantics.

The completion-based rewriting prompt takes the dialogue history, the current query, and the extracted keywords as input, and instructs the model to generate a semantically complete query while preserving core domain terms. The full template is available in the project GitHub repository.

After the rewrite is generated, the system further checks whether the rewritten query contains the pre-extracted keywords or their normalized forms. If obvious omissions are detected, regeneration is triggered with stronger preservation constraints in the prompt, until the requirement is satisfied or the maximum retry limit is reached. Through this process, the output of Stage I is able to recover the information hidden in the dialogue history while maintaining consistency with the original question at the terminology level.

Overall, the first stage of SGQR is a completion-based rewriting process with keyword constraints. It restores the full semantics of the current query by leveraging dialogue history, while controlling terminology drift through keyword preservation and post-generation checking.

3.1.2 Stage II: Entropy-Gated Query Expansion (EGQE)

After completion-based rewriting in Stage I, the query already has a relatively complete semantic expression. However, for technical documents with clear hierarchical structures, such as aviation manuals, semantic completion alone is often insufficient for optimal retrieval. Many passages may be locally similar in meaning while belonging to different systems, sections, or functional paths. As a result, relying only on the query text may still make it difficult to accurately locate the target content. Based on this observation, Stage II of SGQR further expands the query by incorporating structural information from title paths.

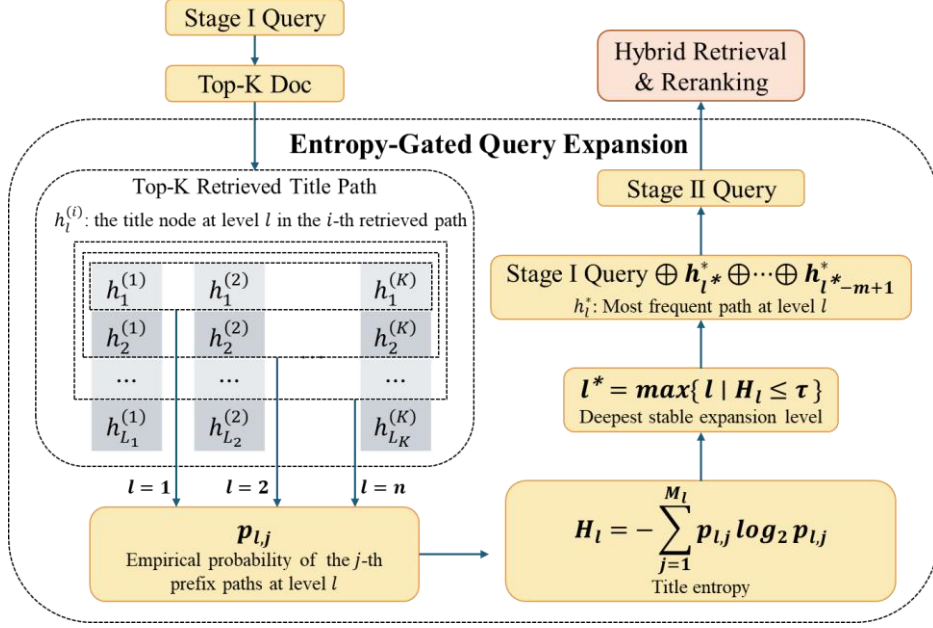


Fig. 2. Entropy-Gated Query Expansion in Stage II of SGQR. The top-K retrieved title paths are decomposed into multi-level nodes, from which the empirical prefix-path distribution is estimated at each level. Prefix entropy is then computed to select the deepest stable expansion level, and the corresponding structural terms are appended to the Stage I query.

The effectiveness of structural information depends on its stability. In general, deeper title nodes provide more precise topical constraints, but they are also more susceptible to noise from initial retrieval. Therefore, the core issue in this stage is how to identify the deepest level of title-path information that is still sufficiently consistent. To this end, we adopt a gating strategy based on prefix-path entropy to uniformly model the structural distribution of initial retrieval results across different levels.

Specifically, we first perform an initial retrieval using the completion-based query q_{comp} obtained in Stage I, yielding the top-K candidate documents together with their corresponding title paths. Let the complete title path of the i -th candidate document be denoted as

$$P^{(i)} = (h_1^{(i)}, h_2^{(i)}, \dots, h_{L_i}^{(i)}), \quad i = 1, \dots, K$$

where $h_l^{(i)}$ denotes the title node of the path at level l . For any level l , the prefix path of the i -th candidate up to level l is defined as

$$\pi_l^{(i)} = (h_1^{(i)}, h_2^{(i)}, \dots, h_l^{(i)})$$

That is, $\pi_l^{(i)}$ represents the structural category of the candidate document at level l .

When l is fixed, the top-K candidates form a set of prefix paths:

$$\{\pi_l^{(1)}, \pi_l^{(2)}, \dots, \pi_l^{(K)}\}$$

We then count the occurrence frequencies of different prefix patterns in this set. Suppose that, at level l , there are M_l distinct prefix-path types, and the j -th prefix appears $c_{l,j}$ times among the top-K candidates. Its empirical probability is defined as

$$p_{l,j} = \frac{c_{l,j}}{K}, j = 1, \dots, M_l$$

Based on this distribution, the prefix-path entropy at level l is defined as

$$H_l = - \sum_{j=1}^{M_l} p_{l,j} \log_2 p_{l,j}$$

This entropy measures the structural dispersion of candidate documents up to level l . If H_l is low, most candidates share the same or similar prefix paths up to that level, indicating that the current query has been relatively stably localized to a certain structural region. In contrast, if H_l is high, the candidates are clearly scattered at that level, and continuing to use structural information from that level or deeper levels may introduce noise.

In this work, the number of initial retrieval candidates is fixed at top- $K = 5$. Under this setting, the entropy of the prefix-path distribution exhibits a relatively clear discrete pattern: when candidates concentrate on a single path or a small number of prefix-path types, the entropy remains low; when they are distributed across three or more prefix types, the entropy increases significantly. Based on this property and validation-set observations, we use a unified entropy threshold $\tau = 1.0$.

Accordingly, the system backtracks from deeper levels to shallower ones and identifies the deepest path level that still satisfies the consistency requirement:

$$l^* = \max\{l \mid H_l \leq \tau\}$$

where l^* denotes the deepest level that can still stably support structural expansion under the current retrieval results. If the prefix paths at deeper levels have already diverged substantially, the system automatically falls back to a shallower level. If deeper levels still maintain high consistency, the system preferentially preserves the deeper structural information to obtain more precise topical constraints.

After determining l^* , the system extracts title terms from the level l^* and its adjacent upper level from the highest-ranked candidate path $\pi_{l^*}^{(1)}$, while filtering out terms that are overly generic or already appear in the current query. These structural terms are then appended to the completion-based query from Stage I, yielding the final expanded query:

$$q_{sgqr} = q_{comp} \oplus t_1 \oplus \dots \oplus t_m$$

where t_i denotes a selected structural title term, and m denotes the number of retained expansion levels. In our experiments, we found that appending too many structural terms after the query tends to dilute the semantics of the query itself and may even degrade retrieval performance. Therefore, the maximum value of m is set to 2, meaning that only the titles at level l^* and level $l^* - 1$ are retained.

In essence, Stage II of SGQR is a controlled structural expansion process based on the consistency of multi-level path distributions. It uses a unified entropy measure to determine the stability of title-structure information across different levels, and preferentially retains the deepest, most precise, and sufficiently stable structural cues. In this way, it strengthens topical constraints while reducing semantic contamination caused by noisy paths.

3.2 Hybrid Retrieval

In complex technical documents such as aviation manuals, user questions and document

content often involve both the need for exact terminology matching and semantic variation in expression. A single retrieval method is usually insufficient to handle both aspects effectively. On the one hand, dense retrieval can capture semantic similarity, but it may be less precise when dealing with domain-specific terms or symbolic expressions. On the other hand, sparse retrieval methods based on term matching, such as BM25 [39], are effective for terminology matching but have limited ability to handle semantic variation caused by differences in expression.

To address this issue, we adopt a hybrid retrieval strategy that combines the strengths of sparse retrieval and dense retrieval to improve overall retrieval performance. Specifically, retrieval is performed at two levels: title and context.

At the context level, where information density is the highest and answer content is directly carried, the context field is treated as the primary retrieval target. For the context field, we use both BM25 and dense retrieval. BM25 is used to capture exact keyword-level matches, especially for domain terms, equipment names, and functional descriptions. Dense retrieval is used to model semantic similarity and compensate for matching failures caused by expression variation. The combination of the two improves the balance between terminology matching and semantic matching.

At the title level, since titles usually reflect the structural semantics and topical categories of the document and are therefore relatively concise, we introduce title-based BM25 retrieval as an auxiliary signal. Title matching helps quickly identify topic-relevant regions, but because title information is limited, it should not be assigned excessive weight.

In implementation, the three types of scores are first normalized and then combined through linear weighting:

$$score = w_t \cdot score_{title}^{bm25} + w_c^{bm25} \cdot score_{context}^{bm25} + w_c^{dense} \cdot score_{context}^{dense}$$

where w_t , w_c^{bm25} , and w_c^{dense} denote the weights of title BM25, context BM25, and context dense retrieval, respectively.

In weight design, we first establish the assignment principle based on task characteristics. Since context relevance should serve as the dominant signal, the overall weight assigned to the context field, combining BM25 and dense retrieval, should be significantly higher than that of the title part. Meanwhile, within the context field, considering the importance of semantic matching in multi-turn dialogue settings, the weight of dense retrieval should be higher than that of BM25, while BM25 is still retained as a critical supplement for terminology alignment. Based on these principles, the weight assignment should satisfy:

$$w_c^{dense} \approx w_c^{bm25} > w_t$$

On this basis, we further tune the weights experimentally and finally adopt the following configuration:

$$w_t = 0.1, w_c^{bm25} = 0.5, w_c^{dense} = 0.4$$

Although the weight tuning process itself is relatively straightforward, it has a stable and significant impact on overall performance. Therefore, in the experimental section, we compare different weight combinations to verify the effectiveness of the selected configuration. The results show that this weight setting achieves good robustness and generalization ability on the current task.

3.3 Answer-Coverage-Based Reranking

After the final retrieval results are obtained, a key question is how to present a limited

number of candidate passages to the large language model in an optimal order so as to maximize answer quality. In traditional retrieval systems, reranking is commonly applied after initial retrieval to further refine the ranking of candidate documents [40]. However, in practical question answering, we observe that relevance alone is insufficient to fully reflect the ability of a document to support answer generation. Some passages may be semantically relevant, yet still lack sufficiently complete information to support answer generation on their own. In contrast, some non-gold passages may contain more direct and complete evidence for answering the question.

Based on this observation, we introduce an Answer-Coverage-Based Reranking strategy from the perspective of improving final generation quality. Specifically, we use a large language model to evaluate each candidate passage and estimate the extent to which it can answer the current question, assigning it a discrete score (e.g., 0–4), referred to as answer coverage (AC). This score does not rely on the gold answer. Instead, it is based on the model’s judgment of information coverage and answer sufficiency, making it more consistent with the actual information needs of the generation process.

After obtaining the AC scores for all candidate passages, we rerank the candidates according to these scores, so that passages with stronger answer support are placed earlier. In essence, this strategy shifts the ranking objective from relevance-oriented ranking to generation-oriented ranking. That is, instead of asking only whether a passage is relevant, the reranking process evaluates whether it is sufficiently informative to support answer generation.

It is worth noting that, in practical generation, we usually provide the top- K candidate passages (e.g., top 2 or top 3) to the large language model jointly. Therefore, the goal of this reranking strategy is not to strictly distinguish the first-ranked passage from the second-ranked one, but rather to concentrate high-quality answer-supporting passages at the top of the ranking, thereby improving overall information utilization efficiency.

Finally, the large language model generates the answer based on the reranked candidate passages. Because the input context now contains higher-quality evidence with stronger answer support, this process can effectively reduce the risk of missing information and erroneous inference during generation.

4 Experiments and Evaluation

4.1 QA Dataset Construction

To evaluate the proposed method under different types of contextual dependence, we construct a multi-turn QA dataset based on aviation manuals. Since aviation manuals are highly structured technical documents, the dataset is designed to preserve document hierarchy and to simulate multiple types of multi-turn interaction patterns.

We first parse each source document into a hierarchical text tree according to its heading structure. The hierarchy is derived from document headings, where higher-level headings form parent nodes and lower-level headings form their descendants. Each node in the text tree is then serialized into a structured unit with two fields, title and context. The title field records the full heading path from the root to the current node, rather than only the local heading, so as to preserve the structural context explicitly. The context field stores the corresponding text passage associated with the node. This representation makes the document structure directly accessible

for retrieval and modeling, while also providing the basic semantic units for subsequent QA data generation.

On top of the text tree representation, we define three dialogue chain sampling strategies to simulate different multi-turn interaction scenarios: Linear Descent Sampling (LDS), Exhaustive Sibling Sampling (ESS), and Random Node Sampling (RNS). LDS follows a parent-child path from higher-level nodes to lower-level nodes, and is used to simulate progressive topic deepening, where later questions usually depend strongly on the preceding context. ESS samples multiple child nodes under the same parent node, and is used to simulate same-level exploration across related subtopics. This setting often introduces referential ambiguity and local topic switching while still remaining under a shared higher-level topic. RNS randomly combines nodes under weak semantic constraints and is used to simulate random topic switching in more open and unstable interaction settings. Compared with LDS and ESS, RNS contains less coherent historical information and is mainly used to evaluate robustness under distracting context. Together, these three strategies provide evaluation scenarios with different levels of contextual dependence and interference.

After sampling node sequences, we use a few-shot prompting approach to generate the multi-turn QA dataset with Gemini-2.5-flash-lite. Separate prompt templates are designed for LDS, ESS, and RNS. The complete prompt templates are available in the project GitHub repository. The resulting dataset statistics are reported in Table 1.

Table 1. Statistics of the generated multi-turn QA dataset

Statistic	LDS	ESS	RNS	Overall
Number of sessions	40	40	40	120
Average number of queries per session	16.3	22.2	13.8	17.4
Total number of queries	652	888	550	2090

4.2 Evaluation Metrics

Cumulative Answer Relevance (CAR).

To evaluate retrieval quality more accurately in multi-turn QA, we introduce a new metric, Cumulative Answer Relevance (CAR). It should be noted that, because the multi-turn QA dataset in this work is constructed by generating a series of dialogue turns from a given document passage, each question is associated with only one annotated gold document ID. However, in real aviation manuals, although only one standard answer source is labeled, many other passages may still be semantically sufficient to answer the question. This makes traditional evaluation methods based solely on whether the unique gold document is retrieved clearly limited. In this task, even if the retrieval results do not hit the gold document, they should still be regarded as high-quality retrieval as long as other retrieved passages can answer the question either independently or jointly.

In addition, in multi-turn question answering over complex technical documents, answers are often distributed across multiple related passages, and the system usually feeds the top retrieved results (e.g., top 2 or top 3) jointly into the generation model for reasoning. Therefore, retrieval quality should not be determined by a single passage alone, but rather by the overall support provided by multiple candidate passages. Although metrics such as Normalized Discounted Cumulative Gain (NDCG) [41] take ranking positions and graded relevance into account, they are still essentially based on the weighted accumulation of single-document

relevance, and do not explicitly model the collaborative effect of multiple evidence passages.

Based on the above analysis, we propose CAR, which models the top- K retrieval results from the perspective of answer coverage so as to capture their cumulative support for the question and provide an evaluation metric that is more consistent with the behavior of practical question answering systems. Specifically, let the AC scores of the top- K retrieved passages be $AC_r \in [0,4]$. We first normalize them into $\bar{AC}_r \in [0,1]$, and then introduce a power-based rank decay to perform weighted aggregation over passages at different ranks, yielding an overall coverage score x . On this basis, we further enhance the result through a nonlinear transformation $1 - (1 - x)^\alpha$, resulting in the final CAR score. The whole process can be written as:

$$CAR@k = 1 - (1 - \frac{\sum_{r=1}^k \bar{AC}_r \cdot r^{-\gamma}}{\sum_{r=1}^k r^{-\gamma}})^\alpha$$

where

$$\bar{AC}_r = \frac{\min(\max(AC_r, 0), 4)}{4}$$

The parameter γ controls the degree of rank decay, while α controls the enhancement strength of multi-evidence accumulation.

By construction, the value of CAR is strictly bounded within $[0,1]$. When none of the retrieved passages can support the answer, CAR is 0; when all passages are fully relevant, CAR is 1. This gives the metric good normalization properties and interpretability. More importantly, CAR produces a score distribution that aligns well with practical intuition in typical retrieval scenarios.

For parameter settings, we use $\gamma = 0.35$, so that the ranking weights decay smoothly, thereby avoiding excessive bias toward the top-1 result while still ensuring that the top-2 and top-3 passages retain substantial influence. This setting is consistent with practical systems in which the first two or three retrieved passages are often jointly used for generation. In addition, we set $\alpha = 2.5$ to amplify high-coverage cases nonlinearly, so that when multiple passages simultaneously provide strong support, the final score can approach 1 more rapidly.

Answer Evaluation.

In multi-turn RAG tasks, improvements in retrieval performance do not necessarily translate directly into improvements in final answer quality. To more directly assess the impact of different methods on generated answers, we further introduce an Answer Evaluation method as a complementary evaluation from the perspective of generation quality. This method uses a large language model as the evaluator to score the generated answers along multiple dimensions.

Specifically, answer quality is evaluated from the following three aspects:

- Correctness: whether the answer is factually accurate, semantically consistent with the question, and free of factual errors;
- Completeness: whether the answer covers the key information required by the question and whether important content is missing;
- Relevance: whether the answer remains tightly focused on the question and avoids irrelevant or redundant information.

Each dimension is scored on a scale from 0 to 5, where 0 indicates very poor performance and 5 indicates excellent performance. The final answer score is obtained by averaging the three dimension scores, yielding a unified overall quality metric. This design provides fine-grained

evaluation while maintaining a clear and consistent overall scoring standard. The prompt templates used for answer-coverage scoring and answer evaluation are available in the project GitHub repository.

Information Hiding Rate (IHR).

After constructing the multi-turn QA dataset, an important question is whether the dataset can realistically reflect the challenges of multi-turn RAG. An ideal multi-turn QA dataset should not only contain semantically related question sequences, but should also capture the characteristic phenomenon that, as the dialogue proceeds, the current query becomes increasingly dependent on dialogue history through references, omissions, and semantic continuation. If all questions can be understood independently of context, the dataset would be closer to single-turn question answering and would fail to reflect the actual difficulty of multi-turn dialogue modeling.

To quantify the degree to which the current query depends on historical context, we introduce the Information Hiding Rate (IHR). Intuitively, if a query can still express relatively complete semantics without historical information, its information hiding rate is low. Conversely, if the query contains a large amount of implicit information that must be recovered from context, its information hiding rate is high. In essence, IHR measures the degree of semantic loss caused by removing contextual information from the query.

Specifically, for each query, we construct its original query representation q and an expanded representation q_h that incorporates dialogue history. Here, q_h is obtained by concatenating the current query with its corresponding dialogue history, and is used to approximate the semantic expression of the query in its full context. In practice, the current question in a dialogue is usually not strongly related to turns that are too far back. For later turns, concatenating the entire dialogue history may introduce excessive noise. Therefore, we concatenate at most the two turns immediately preceding the current query. We then use the pretrained text representation model BGE-M3 [42] to encode q and q_h , and define the Information Hiding Rate based on the cosine similarity between their vector representations as

$$\text{IHR}(q) = 1 - \cos(f(q), f(q_h))$$

where $f(\cdot)$ denotes the text embedding function. Under this definition, a lower cosine similarity indicates a larger deviation between the original query and its full-context semantic expression after context is removed. This in turn means that more information is hidden in the dialogue history, corresponding to a higher IHR.

For IHR, we do not explicitly construct a true fully specified query. Instead, we use the concatenation of the dialogue history and the current query as its approximate representation. This design avoids dependence on additional generative models or manual annotation, thereby reducing potential model bias and improving the objectivity and reproducibility of the metric. Although this approximation is not fully equivalent to a truly complete query expression, it can still stably capture the information-hiding phenomenon at the dataset level, and is therefore suitable for analyzing dataset difficulty.

4.3 Dataset Difficulty Analysis

In the experimental setup, we analyze the distribution of IHR on the three QA datasets constructed using LDS, ESS, and RNS, respectively, and report statistics such as the mean and median to examine the degree of contextual dependence under different dialogue scenarios. The

results are shown in Fig. 3.

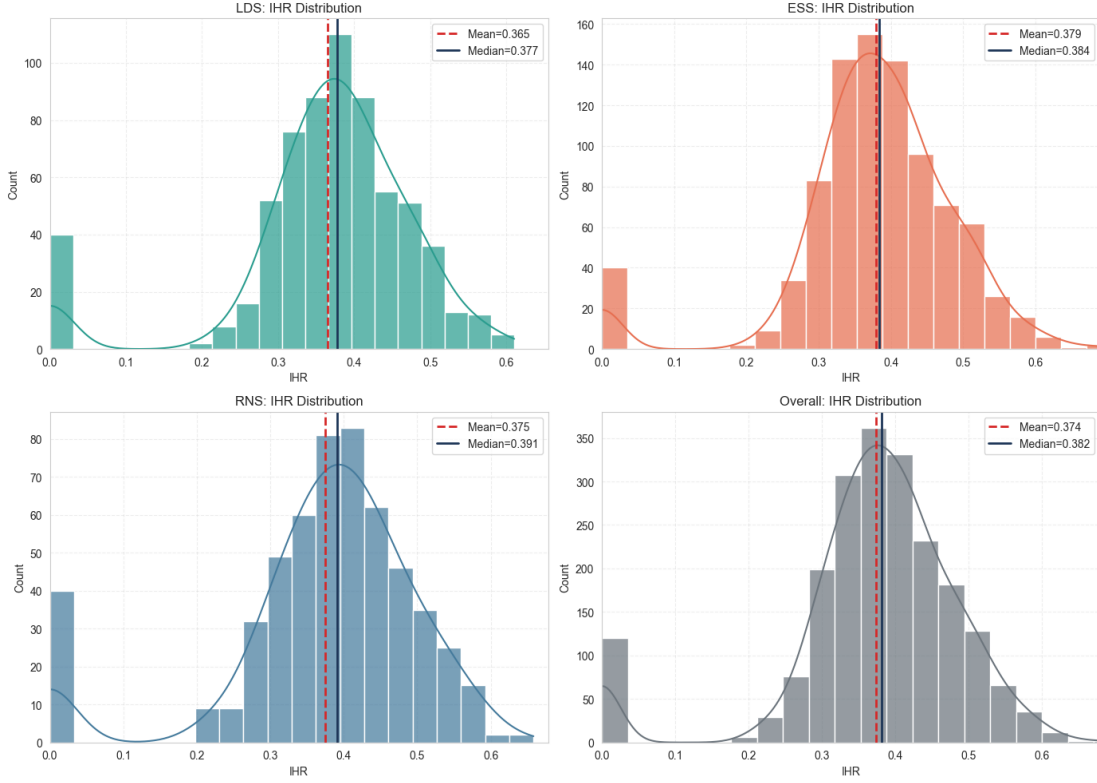


Fig. 3. Distribution of IHR

From the overall distribution, except for the first query in each dialogue session, which is identical to itself and therefore exhibits no hidden information, the Information Hiding Rate shows a relatively continuous distribution with substantial variation across the dataset. This indicates that the incompleteness of query expression is a common phenomenon in multi-turn QA settings, while also varying considerably across samples. In other words, the dataset contains not only queries with relatively complete semantics, but also information-insufficient queries that rely heavily on dialogue history.

On this basis, we further analyze the relationship between IHR and the performance of the baseline method. Specifically, using the original query without rewriting or expansion, we examine the variation trends of $MRR@5$ and $CAR@5$ across different IHR intervals. The results are shown in Fig. 4 and Fig. 5.

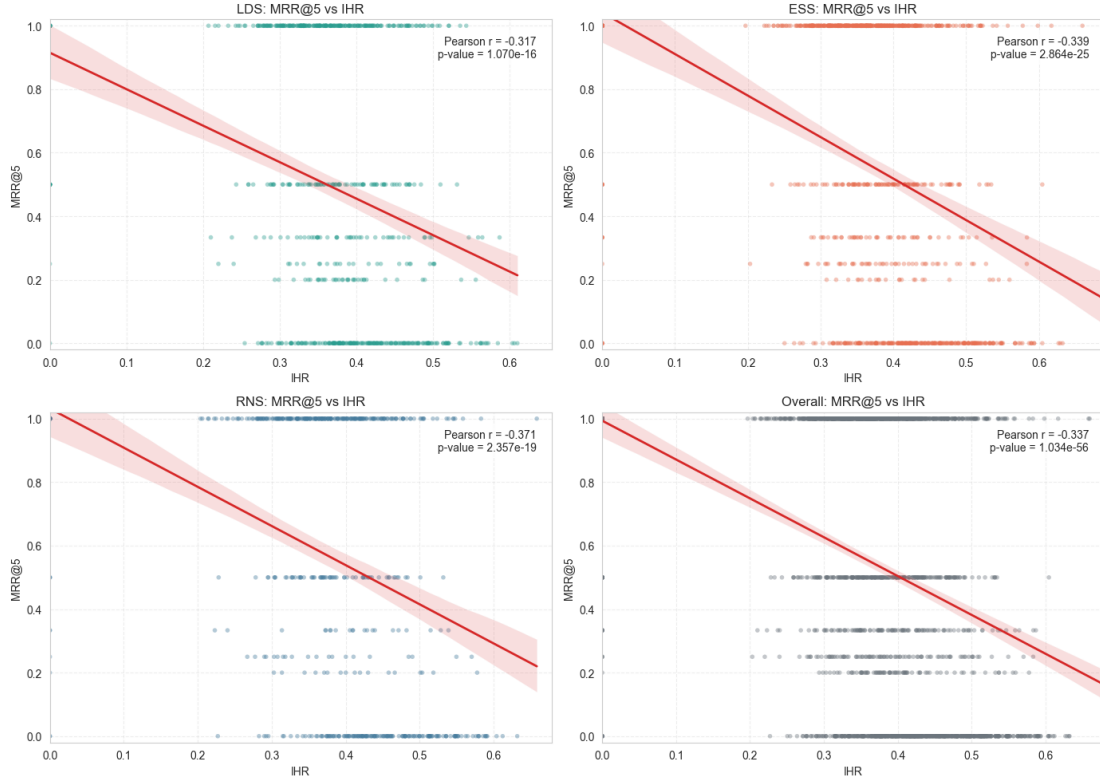


Fig. 4. Correlation Analysis: MRR@5 vs IHR



Fig. 5. Correlation Analysis: CAR@5 vs IHR

Overall, IHR exhibits a relatively continuous trend across the dataset, indicating that multi-turn QA queries differ substantially in terms of information completeness, with different samples corresponding to different degrees of semantic omission. This suggests that the

constructed dataset does not consist of samples with a single difficulty level, but instead forms a difficulty distribution with clear gradation in terms of information hiding, thereby providing a multi-level test environment for multi-turn RAG.

From a statistical perspective, the relationship between IHR and retrieval performance shows that, as IHR increases, system performance generally declines. Moreover, the correlation analysis indicates a significant negative correlation between IHR and the performance metrics, with the p -value being far smaller than 0.05, which demonstrates that the correlation is statistically significant. This result suggests a stable relationship between the degree of hidden information and retrieval performance: the more information is hidden in the dialogue history, the more likely retrieval performance is to degrade when the system relies only on the current query.

Overall, IHR not only captures the information-hiding phenomenon in multi-turn QA at the dataset level, but also reflects its stable relationship with retrieval performance. Therefore, the constructed multi-turn QA dataset is sufficiently challenging in terms of query completeness and can serve as a meaningful benchmark for subsequent method evaluation.

4.4 Ablation Study

To systematically evaluate the role of each module in the multi-turn RAG task, we conduct a progressively enhanced ablation study to compare retrieval and generation performance under different method configurations. Specifically, we start from the original query and then incrementally add query rewriting, structural expansion, and reranking, resulting in the following method settings:

- Original Query: directly uses the user query in the current turn for retrieval, without incorporating dialogue history or any additional processing, and serves as the basic baseline;
- Regular Query Rewriting (Regular QR): introduces completion-based query rewriting with keyword preservation on top of the original query;
- QR + EGQE: further introduces structural query expansion based on Regular QR, that is, the query is structurally enhanced using title-path information from the initial retrieval results;
- QR + EGQE + Reranking: further introduces Answer-Coverage-Based Reranking on top of structure-guided query expansion, which corresponds to the full method proposed in this paper.

In terms of experimental data, we evaluate all methods on the QA datasets constructed using the three dialogue chain sampling strategies, namely LDS, ESS, and RNS. The retrieval-level evaluation metrics are $MRR@5$ and $CAR@5$, while the generation-level metric is Answer Evaluation.

In implementation, all experiments in this section use models from the Qwen3 [43] series. Specifically, Qwen3-8b is used for the query rewriting component in SGQR, while Qwen3-32b is used for AC scoring and Answer Evaluation. The smaller model is adopted for query rewriting because this component may need lightweight deployment in practical applications. By contrast, the larger and stronger model is used for evaluation in order to ensure fairness and stability.

In addition, to ensure the comparability of the experimental results, all methods are evaluated under the same dataset split and retrieval settings, so that performance differences

across configurations can be attributed only to the methods themselves.

Table 2. Results of the ablation study

Scenario	Aspect	Metric	Original Query	Regular QR	QR+ EGQE	QR+ EGQE+ Reranking
LDS	Retrieval	MRR@5	0.4954	0.6653	0.6958	0.8126
		CAR@5	0.8419	0.8653	0.8742	0.8897
	Answer	Correctness	3.9340	4.3451	4.3865	4.3880
		Completeness	3.5261	3.9525	3.9785	3.9893
		Relevance	4.1840	4.5276	4.5567	4.5567
		Final score	3.8388	4.2350	4.2692	4.2738
ESS	Retrieval	MRR@5	0.5463	0.7217	0.7356	0.8421
		CAR@5	0.7714	0.8285	0.8340	0.8503
	Answer	Correctness	3.7399	4.0957	4.1408	4.1453
		Completeness	3.3592	3.7354	3.7680	3.7725
		Relevance	4.0146	4.3615	4.3908	4.4000
		Final score	3.6479	4.0095	4.0478	4.0530
RNS	Retrieval	MRR@5	0.5691	0.7900	0.8013	0.8505
		CAR@5	0.8107	0.8230	0.8294	0.8438
	Answer	Correctness	3.7418	4.1855	4.2509	4.2436
		Completeness	3.4127	3.8564	3.8982	3.8927
		Relevance	4.0073	4.3945	4.4436	4.4364
		Final score	3.6665	4.1016	4.1564	4.1497
Overall	Retrieval	MRR@5	0.5364	0.7221	0.7405	0.8351
		CAR@5	0.8037	0.8380	0.8453	0.8609
	Answer	Correctness	3.8010	4.1971	4.2464	4.2469
		Completeness	3.4254	3.8349	3.8679	3.8718
		Relevance	4.0656	4.4220	4.4565	4.4584
		Final score	3.7092	4.1041	4.1455	4.1473

The ablation results show that the performance gains of the proposed method do not come from any single module, but rather from the cumulative contribution of semantic completion, structural expansion, and Answer-Coverage-Based Reranking. Among them, completion-based query rewriting brings the largest performance improvement, indicating that incomplete query semantics remain the primary bottleneck affecting retrieval effectiveness in multi-turn dialogue settings. Compared with directly using the original query, Regular QR yields clear improvements across all three datasets as well as in the overall results. This shows that, as long as the omitted or implicit key information in the current question can be recovered in a reasonably stable manner, the retrieval system can more accurately align with the user’s true intent. Therefore, in this task, query rewriting is not merely an additional optimization step, but a fundamental component for improving the performance of multi-turn RAG.

On this basis, EGQE brings further stable but relatively limited gains. This suggests that structural information can indeed provide additional constraints for retrieval, but its role is closer to that of a supplementary topical calibration signal rather than the dominant source of performance improvement. In other words, once the query itself already has a relatively

complete semantic expression, introducing title hierarchy and structural path information can further narrow the retrieval space and strengthen topical consistency. However, if query understanding itself is inaccurate, structural information alone is insufficient to fundamentally improve the results. This phenomenon also indicates that the value of structural information depends on the correctness of the underlying query semantics, which explains why its gain is usually smaller than that of completion-based rewriting.

The full method achieves the best results after introducing Answer-Coverage-Based Reranking, indicating that the objective of retrieval in multi-turn RAG is not only to find documents that are relevant, but also to find evidence that can effectively support answer generation. Unlike the first two modules, the reranking stage does not improve performance by reinterpreting the query, but by reorganizing candidate evidence so that high-quality passages are more concentrated at the top of the ranking. Therefore, what it optimizes is not whether the target appears in the candidate set, but whether the candidates are truly useful for answering the question. This is particularly evident in CAR and Answer Evaluation: the full method improves not only ranking quality, but also the correctness, completeness, and relevance of the final answers. This shows that the advantage of the proposed method is not limited to retrieval itself, but can also be transferred to the downstream generation stage, thereby demonstrating stronger task-level consistency.

Across the three dialogue scenarios, the improvement trends brought by each module are generally stable, but the absolute performance differs clearly: LDS performs best, followed by RNS, while ESS is relatively the weakest. This suggests that the mere presence of dialogue history does not necessarily help multi-turn RAG. What matters is whether the history remains sufficiently consistent with the current question. In LDS, the topic deepens progressively along a structural path, so the historical information has strong semantic continuity with the current query, making both query completion and structural expansion more effective. In ESS, although the dialogue remains under the same parent topic, lateral switching across sibling subtopics is more likely to introduce local drift, so the historical information is relevant but not fully aligned with the current question, and is therefore more likely to interfere with query understanding. In RNS, topic switching is more obvious, and although there is less useful historical information, misleading context is also reduced. As a result, its overall performance is better than that of ESS. This indicates that the key challenge of multi-turn RAG is not simply whether historical information exists, but whether the system can effectively distinguish which parts of the history should be inherited and which should be suppressed.

Overall, the ablation study reveals a clear conclusion: the performance improvement of the proposed method is established through a layer-by-layer optimization process. Completion-based rewriting first addresses the problem that the current question is semantically incomplete. Structural expansion then further addresses how the query can be aligned more precisely with document topics. Finally, Answer-Coverage-Based Reranking determines which candidates are most worth passing to the generation stage. These three components operate at the levels of query formulation, retrieval matching, and evidence organization, respectively, and together explain the effectiveness of the full method. This also shows that the effectiveness of SGQR-RAG does not come from any isolated trick, but from a layered treatment of the key bottlenecks in multi-turn RAG.

4.5 Model Generalization Analysis

To evaluate the applicability and stability of the proposed method across different large language models, we further conduct cross-model experiments. This experiment examines whether SGQR-RAG can consistently outperform the baseline Regular QR across different model settings. To this end, we select several mainstream open-source instruction-tuned models, including Qwen3, LLaMA3.1 [44], Ministral3 [45], and Gemma3 [46]. These models cover different model families and different capability levels, so as to reduce the influence of bias from any single model as much as possible.

In this experiment, only the large language model used for query rewriting is replaced, while the scoring process still uses Qwen3-32b. In addition, as the results in the previous subsection show that better retrieval performance usually corresponds to better answer generation performance, we compare only retrieval effectiveness here.

Table 3. Retrieval performance with different query rewriting models

LLM	Method	MRR@5	CAR@5
Qwen3-1.7b	Regular QR	0.7529	0.8179
	SGQR-RAG	0.8171	0.8346
Qwen3-8b	Regular QR	0.7221	0.8380
	SGQR-RAG	0.8351	0.8609
Qwen3-32b	Regular QR	0.7524	0.8498
	SGQR-RAG	0.8528	0.8690
LLaMA3.1-8b-Instruct	Regular QR	0.6550	0.7913
	SGQR-RAG	0.7812	0.8345
LLaMA3.1-70b-Instruct	Regular QR	0.7477	0.8283
	SGQR-RAG	0.8522	0.8632
Ministral3-8b-2512	Regular QR	0.7106	0.7856
	SGQR-RAG	0.7873	0.8344
Ministral3-14b-2512	Regular QR	0.7055	0.8021
	SGQR-RAG	0.8346	0.8515
Gemma3-4b	Regular QR	0.6893	0.7916
	SGQR-RAG	0.8185	0.8345
Gemma3-27b	Regular QR	0.7832	0.8416
	SGQR-RAG	0.8475	0.8577

From the overall results, SGQR-RAG outperforms Regular QR on all tested models, and shows consistent improvements on both MRR@5 and CAR@5. This indicates that the benefits of the proposed method are stable rather than being limited to a specific model type or an isolated configuration. In other words, the improvement brought by SGQR-RAG does not depend on a particular model architecture, but can serve as a general strategy for query enhancement and retrieval optimization across different large language models.

This consistent improvement is reflected at two levels. On the one hand, the gain in MRR@5 shows that the proposed method is more effective at ranking key evidence documents near the top, thereby improving retrieval ranking quality. On the other hand, the simultaneous improvement in CAR@5 indicates that SGQR-RAG not only improves the ability to retrieve the most relevant document, but also enhances the effective information coverage within the top- K candidates that can be used to answer the question. Therefore, the advantage of the

method is not limited to relevance ranking in the conventional sense, but is further reflected in evidence organization that is more valuable for downstream generation.

In addition, the absolute performance varies across models, and stronger models generally achieve higher overall scores, indicating that the inherent capability of the underlying model still affects the upper bound of performance. However, this is not the main focus of the present experiment. More importantly, regardless of whether the model itself is relatively strong or weak, SGQR-RAG consistently brings positive gains over Regular QR.

Overall, the cross-model experiments verify that SGQR-RAG has good generalization ability and model adaptability. Its effectiveness does not depend on a specific model family or parameter scale, but can stably improve retrieval quality and evidence usefulness across multiple mainstream models, thus demonstrating strong practical value.

5 Conclusion

This paper focuses on multi-turn question answering over aviation manuals and proposes SGQR-RAG, a structure-guided retrieval framework designed to address incomplete query semantics, terminology drift, and the ineffective use of document structural information. First, we construct a multi-turn QA dataset for aviation manuals and analyze the contextual dependence of multi-turn queries at the data level. On this basis, we propose a two-stage query rewriting method, SGQR. In Stage I, the complete semantics of the current query are recovered through completion-based rewriting with keyword preservation constraints. In Stage II, high-confidence structural information is introduced in a controlled manner through gated structural expansion based on the multi-level prefix-path entropy of title paths. Furthermore, we combine Hybrid Retrieval with Answer-Coverage-Based Reranking to optimize evidence selection from the perspective of retrieval-generation collaboration. Experimental results show that the proposed method achieves consistent improvements across different dialogue scenarios and multiple open-source large language models, and overall outperforms baseline methods in retrieval ranking, evidence coverage, and answer quality, thereby verifying its effectiveness for multi-turn question answering over aviation manuals.

Despite these contributions, this work still has several limitations. First, some modules in the current framework still rely on the generation and judgment capabilities of large language models, and their effectiveness may therefore be affected by differences in the capability of the underlying models. Second, the structural expansion stage is currently gated mainly by the consistency of title paths, and has not yet incorporated finer-grained semantic relations or internal document structural information. Third, the experiments in this paper are conducted mainly in the aviation manual scenario. Although this is sufficient to verify the effectiveness of the proposed method on complex professional documents, it has not yet been systematically evaluated on classic multi-turn dialogue benchmarks such as QuAC, CANARD, and TREC CAsT. Therefore, the current conclusions mainly reflect the applicability of the method to multi-turn question answering in the aviation domain, while its performance in general conversational question answering and conversational retrieval settings still requires further validation.

Future work will proceed in three directions. First, we will explore lighter-weight and more stable query rewriting and reranking mechanisms, so as to reduce dependence on the capability of large language models. Second, we will incorporate richer document structures and domain

knowledge representations to further improve the precision of structural information utilization. Third, we will extend the framework to classic multi-turn dialogue benchmarks and a wider range of technical document scenarios, and systematically evaluate its generalization ability across datasets and domains.

Declaration of Competing Interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

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Code and Data Availability

The code, prompts, and experimental resources used in this study are available at the project GitHub repository: <https://github.com/Mephisto0518/SGQR-RAG>.

References

- [1] Endsley M R .Toward a Theory of Situation Awareness in Dynamic Systems[J].Human Factors: The Journal of the Human Factors and Ergonomics Society, 1995.DOI:10.1518/001872095779049543.
- [2] Young M S , Brookhuis K A , Wickens C D ,et al.State of science: mental workload in ergonomics[J].Ergonomics, 2015, 58(1):1-17.DOI:10.1080/00140139.2014.956151.
- [3] Hamed G T ,Nicholas W ,Jessica V , et al.Using machine learning methods and EEG to discriminate aircraft pilot cognitive workload during flight.[J].Scientific reports,2023,13(1):2507-2507.DOI:10.1038/S41598-023-29647-0.
- [4] Analysis L .Introduction to Information Retrieval[M].Cambridge University Press,2008.
- [5] Lewis P , Perez E , Piktus A ,et al.Retrieval-Augmented Generation for Knowledge-Intensive NLP Tasks[J]. 2020.DOI:10.48550/arXiv.2005.11401.
- [6] Borgeaud S , Mensch A , Hoffmann J ,et al.Improving language models by retrieving from trillions of tokens[J].arXiv e-prints, 2021.DOI:10.48550/arXiv.2112.04426.
- [7] Vakulenko, S., Longpre, S., Tu, Z. & Anantha, R. Question Rewriting for Conversational Question Answering. Proceedings of the 14th ACM International Conference on Web Search and Data Mining 355–363 (2021) doi:10.1145/3437963.3441748.
- [8] Elgohary, A., Peskov, D., & Boyd-Graber, J. (2019). Can You Unpack That? Learning to Rewrite Questions-in-Context. In K. Inui, J. Jiang, V. Ng, & X. Wan (Eds.), Proceedings of the 2019 Conference on Empirical Methods in Natural Language Processing and the 9th International Joint Conference on Natural Language Processing (EMNLP-IJCNLP)(pp. 5917–5923). Association for Computational Linguistics.
<https://doi.org/10.18653/v1/d19-1605>
- [9] Nogueira R , Cho K .Passage Re-ranking with BERT[J]. 2019.DOI:10.48550/arXiv.1901.04085.
- [10] Karpukhin V , Ouz B , Min S ,et al.Dense Passage Retrieval for Open-Domain Question Answering[J]. 2020.DOI:10.18653/v1/2020.emnlp-main.550.
- [11] Arnold A ,Gérard Dupont,Félix Furger,et al.A question-answering system for aircraft pilots' documentation[J]. 2020.DOI:10.48550/arXiv.2011.13284.
- [12] Agarwal A , Gite R , Laddha S ,et al.Knowledge Graph - Deep Learning: A Case Study in Question Answering in Aviation Safety Domain[J]. 2022.
- [13] Wang L , Chou J , Tien A ,et al.AviationGPT: A Large Language Model for the Aviation Domain[J].AIAA AVIATION FORUM AND ASCEND 2024, 2024.DOI:10.2514/6.2024-4250.
- [14] Zhang Z , Feng S , Yang T ,et al.AviationCopilot: Building a reliable LLM-based Aviation Copilot inspired by human pilot training[J].Advanced Engineering Informatics, 2026, 69(pa):103806.DOI:10.1016/j.aei.2025.103806.

- [15] Chi G , Tang X , Yung K L ,et al.Exploring Research on the Construction and Application of Knowledge Graphs for Aircraft Fault Diagnosis[J].Sensors (Basel, Switzerland), 2023, 23(11):5295-5295.DOI:10.3390/s23115295.
- [16] Ai K , Karr Jr J A , Jiang M ,et al.KEO: Knowledge Extraction on OMIn via Knowledge Graphs and RAG for Safety-Critical Aviation Maintenance[J]. 2025.
- [17] Li L ,Haruna A ,Ying W , et al.Knowledge graph-driven fault diagnosis for aviation equipment: Integrating improved joint extraction with large language model[J].Journal of Industrial Information Integration,2026,50101039-101039.DOI:10.1016/J.JII.2025.101039.
- [18] Xie X , Tang X , Gu S ,et al.An intelligent guided troubleshooting method for aircraft based on HybirdRAG[J].Scientific Reports[2026-03-24].DOI:10.1038/s41598-025-02643-2.
- [19] Ren T , Zhang Z , Jia B ,et al.Retrieval-Augmented Generation-aided causal identification of aviation accidents: A large language model methodology[J].Expert Systems with Application, 2025(Jun.):278.DOI:10.1016/j.eswa.2025.127306.
- [20] Xiang X ,Chen X ,Yang J .Knowledge Graph-Based Causal Analysis of Aviation Accidents: A Hybrid Approach Integrating Retrieval-Augmented Generation and Prompt Engineering[J].Aerospace,2025,13(1):16-16.DOI:10.3390/AEROSPACE13010016.
- [21] Xiong X , Cai H , Yu H ,et al.DR-RAG: Domain-Rule-based Retrieval-Augmented Generation for aviation digital model design[J].Advanced Engineering Informatics, 2025, 68(pb):103688.DOI:10.1016/j.aei.2025.103688.
- [22] Silva E A D C , Marsh R , Yong H ,et al.Retrieval-Augmented Generation and In-Context Prompted Large Language Models in Aircraft Engineering[J].AIAA SCITECH 2025 Forum, 2025.DOI:10.2514/6.2025-0700.
- [23] Feng S , Wan H , Gunasekara C ,et al.doc2dial: A Goal-Oriented Document-Grounded Dialogue Dataset[J]. 2020.DOI:10.18653/v1/2020.emnlp-main.652.
- [24] Feng S , Patel S S , Wan H ,et al.MultiDoc2Dial: Modeling Dialogues Grounded in Multiple Documents[J]. 2021.DOI:10.48550/arXiv.2109.12595.
- [25] Ma L , Zhang W N , Li M ,et al.A Survey of Document Grounded Dialogue Systems (DGDS)[J]. 2020.DOI:10.48550/arXiv.2004.13818.
- [26] Feng S. DialDoc 2021 Shared Task: Goal-Oriented Document-grounded Dialogue Modeling. In: Feng S, Reddy S, Alikhani M, He H, Ji Y, Iyyer M, et al., eds. Proceedings of the 1st Workshop on Document-grounded Dialogue and Conversational Question Answering (DialDoc 2021). Online: Association for Computational Linguistics; 2021. p. 1–7. <https://doi.org/10.18653/v1/2021.dialdoc-1.1>
- [27] Daheim N , Thulke D , Dugast C ,et al.Cascaded Span Extraction and Response Generation for Document-Grounded Dialog[J]. 2021.DOI:10.18653/v1/2021.dialdoc-1.8.

- [28] Gao C , Zhang W , Lam W .UniGDD: A Unified Generative Framework for Goal-Oriented Document-Grounded Dialogue[J].Proceedings of the 60th Annual Meeting of the Association for Computational Linguistics (Volume 2: Short Papers), 2022:599-605.DOI:10.18653/v1/2022.acl-short.66.
- [29] Kim B , Lee D , Kim S ,et al.Document-Grounded Goal-Oriented Dialogue Systems on Pre-Trained Language Model with Diverse Input Representation[J]. 2021.DOI:10.18653/v1/2021.dialdoc-1.12.
- [30] Roy N, Ribeiro LFR, Blloshmi R, Small K. Learning When to Retrieve, What to Rewrite, and How to Respond in Conversational QA. In: Al-Onaizan Y, Bansal M, Chen Y-N, eds. Findings of the Association for Computational Linguistics: EMNLP 2024. Miami, Florida, USA: Association for Computational Linguistics; 2024. p. 10604–10625. <https://doi.org/10.18653/v1/2024.findings-emnlp.622>
- [31] Anantha R, Vakulenko S, Tu Z, Longpre S, Pulman S, Chappidi S. Open-Domain Question Answering Goes Conversational via Question Rewriting. In: Toutanova K, Rumshisky A, Zettlemoyer L, Hakkani-Tur D, Beltagy I, Bethard S, et al., eds. Proceedings of the 2021 Conference of the North American Chapter of the Association for Computational Linguistics: Human Language Technologies. Online: Association for Computational Linguistics; 2021. p. 520–534. <https://doi.org/10.18653/v1/2021.naacl-main.44>
- [32] Wu Z, Luan Y, Rashkin H, Reitter D, Hajishirzi H, Ostendorf M, Tomar GS. CONQRR: Conversational Query Rewriting for Retrieval with Reinforcement Learning. In: Goldberg Y, Kozareva Z, Zhang Y, eds. Proceedings of the 2022 Conference on Empirical Methods in Natural Language Processing. Abu Dhabi, United Arab Emirates: Association for Computational Linguistics; 2022. p. 10000–10014. <https://doi.org/10.18653/v1/2022.emnlp-main.679>
- [33] Qian H, Dou Z. Explicit Query Rewriting for Conversational Dense Retrieval. In: Goldberg Y, Kozareva Z, Zhang Y, eds. Proceedings of the 2022 Conference on Empirical Methods in Natural Language Processing. Abu Dhabi, United Arab Emirates: Association for Computational Linguistics; 2022. p. 4725–4737. <https://doi.org/10.18653/v1/2022.emnlp-main.311>
- [34] Mao K, Dou Z, Liu B, Qian H, Mo F, Wu X, et al. Search-Oriented Conversational Query Editing. In: Rogers A, Boyd-Graber J, Okazaki N, eds. Findings of the Association for Computational Linguistics: ACL 2023. Toronto, Canada: Association for Computational Linguistics; 2023. p. 4160–4172. <https://doi.org/10.18653/v1/2023.findings-acl.256>
- [35] Mo F, Ghaddar A, Mao K, Rezagholizadeh M, Chen B, Liu Q, et al. CHIQ: Contextual History Enhancement for Improving Query Rewriting in Conversational Search. In: Al-Onaizan Y, Bansal M, Chen Y-N, eds. Proceedings of the 2024 Conference on Empirical

Methods in Natural Language Processing. Miami, Florida, USA: Association for Computational Linguistics; 2024. p. 2253–2268.

<https://doi.org/10.18653/v1/2024.emnlp-main.135>

- [36] Zhang T, Li K, Luo H, Wu X, Glass JR, Meng HM. Adaptive Query Rewriting: Aligning Rewriters through Marginal Probability of Conversational Answers. In: Al-Onaizan Y, Bansal M, Chen Y-N, eds. Proceedings of the 2024 Conference on Empirical Methods in Natural Language Processing. Miami, Florida, USA: Association for Computational Linguistics; 2024. p. 13444–13461. <https://doi.org/10.18653/v1/2024.emnlp-main.746>
- [37] Mao S, Jiang Y, Chen B, Li X, Wang P, Wang X, Xie P, Huang F, Chen H, Zhang N. RaFe: Ranking Feedback Improves Query Rewriting for RAG. In: Al-Onaizan Y, Bansal M, Chen Y-N, eds. Findings of the Association for Computational Linguistics: EMNLP 2024. Miami, Florida, USA: Association for Computational Linguistics; 2024. p. 884–901. <https://doi.org/10.18653/v/2024.findings-emnlp.49>
- [38] Grootendorst M. KeyBERT: Minimal keyword extraction with BERT [Computer software]. Zenodo, 2020. <https://doi.org/10.5281/zenodo.4461265> (Version v0.9.0)
- [39] Robertson S , Zaragoza H .The Probabilistic Relevance Framework: BM25 and Beyond[J].Foundations & Trends in Information Retrieval, 2009, 3(4):333-389.DOI:10.1561/15000000019.
- [40] Nogueira R ,Cho K .Passage Re-ranking with BERT.[J].CoRR,2019,abs/1901.04085
- [41] Jaervelin K , Kekaelaenen J .Cumulated gain-based evaluation of IR techniques[J].Acm Transactions on Information Systems, 2002, 20(4):422-446.DOI:10.1145/582415.582418.
- [42] Chen J , Xiao S , Zhang P ,et al.M3-Embedding: Multi-Linguality, Multi-Functionality, Multi-Granularity Text Embeddings Through Self-Knowledge Distillation[J].Findings of the Association for Computational Linguistics ACL 2024, 2024:2318-2335.DOI:10.18653/v1/2024.findings-acl.137.
- [43] Yang, A., Li, A., Yang, B., et al. Qwen3 Technical Report. arXiv preprint arXiv:2505.09388, 2025. DOI: 10.48550/arXiv.2505.09388.
- [44] Grattafiori, A., Dubey, A., Jauhri, A., et al. The Llama 3 Herd of Models. arXiv preprint arXiv:2407.21783, 2024. DOI: 10.48550/arXiv.2407.21783.
- [45] Liu, A. H., Khandelwal, K., Subramanian, S., et al. Ministral 3. arXiv preprint arXiv:2601.08584, 2026. DOI: 10.48550/arXiv.2601.08584.
- [46] Gemma Team. Gemma 3 Technical Report. arXiv preprint arXiv:2503.19786, 2025. DOI: 10.48550/arXiv.2503.19786.

Declaration of interests

☒The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

☐The authors declare the following financial interests/personal relationships which may be considered as potential competing interests: